Ethanol Organosolv Lignin Depolymerisation with Hydrogen over a Pd/C Catalyst

Aalto University
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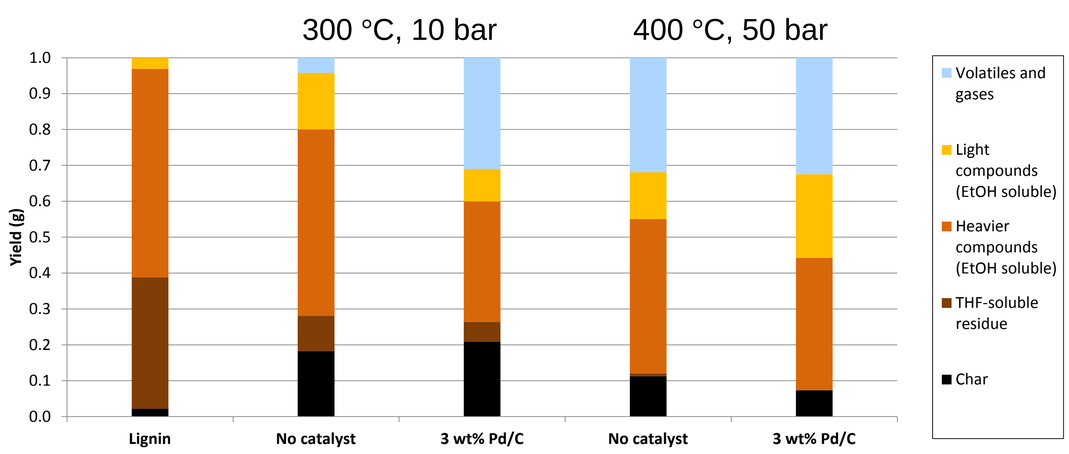
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Introduction

In the bio-based economy, lignin is the most potential resource for aromatic chemicals and fuels, and catalysis is regarded to be the key technology for its valorisation. [1] In addition, hydrogenolysis is considered to be the most promising method to produce phenols from lignin. [2] In this work, lignin depolymerisation was studied in ethanol with hydrogen and a 3 wt-% Pd/C catalyst. Ethanol organosolv lignin (EOL) was selected as a feedstock because it is a sulphur-free, low-molecular-weight product of high purity. [3]

Results

Effect of the Catalyst



- Catalyst increased clearly the lignin depolymerisation, however, much more volatile/gaseous compounds was formed (300 °C, 10 bar)
- Effect on the light compound (monomers and dimers) and char formation depends on the conditions, but catalyst modifies light compound distribution

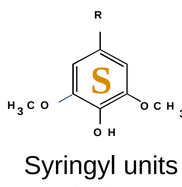
EOL Characterisation Results

GPC

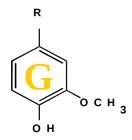
EOL Fraction	Mn (g/mol)	Mw (g/mol)	PDI
EOL	726	1799	2.48
EOL EtOH Soluble	578	932	1.61
EOL THF Soluble (Residue)	1225	3005	2.45

Elemental Analysis

Element	Amount (wt%)
Carbon	61.1
Oxygen	31.8
Hydrogen	6.3
Nitrogen	0.3
Sulfur	0.4



Syringyl units **67 %**

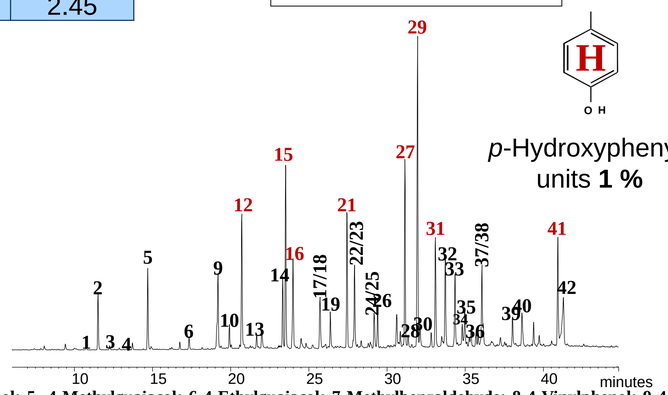


Guaiacyl units **32 %**

Pyrolysis-GC/MS
600 °C, 2 s



p-Hydroxyphenyl units **1 %**



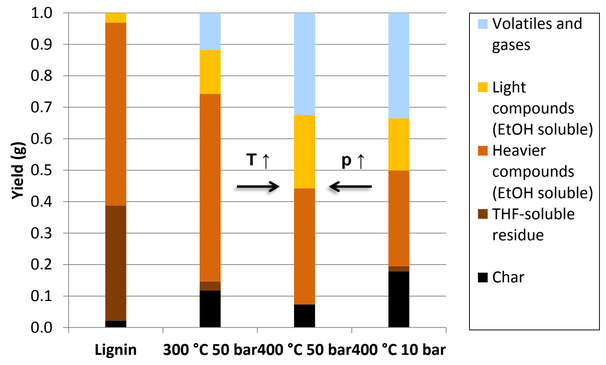
Peak identities:
1 Phenol; 2 Guaiacol; 3 2-Methylphenol; 4 4-Methylphenol; 5 4-Methylguaiacol; 6 4-Ethylguaiacol; 7 Methylbenzaldehyde; 8 4-Vinylphenol; 9 4-Vinylguaiacol; 10 Eugenol; 11 Catechol; 12 Syringol; 13 *cis*-Isoeugenol; 14 *trans*-Isoeugenol; 15 4-Methylsyringol; 16 Vanillin; 17 4-Ethylsyringol; 18 Homovanillin; 19 Acetoguaiacone; 20 Hydroxybenzaldehyde; 21 4-Vinylsyringol; 22 Guaiacylacetone; 23 4-Allylsyringol; 24 4-(oxy-allyl)guaiacol; 25 4-(1-Hydroxyprop-2-enyl)guaiacol; 26 *cis*-Propenylsyringol; 27 *trans*-Propenylsyringol; 28 Dihydroconiferyl alcohol; 29 Syringaldehyde; 30 *cis*-Coniferyl alcohol; 31 Homosyringaldehyde; 32 Acetosyringone; 33 *trans*-Coniferyl alcohol; 34 Syringylacetone; 35 Coniferaldehyde; 36 Propiosyringol; 37 4-(oxy-allyl)syringol; 38 4-(1-Hydroxy-prop-2-enyl)syringol; 39 Dihydrosinapyl alcohol; 40 *cis*-Sinapyl alcohol 41 *trans*-Sinapyl alcohol; 42 Sinapaldehyde

Experimental

- 100 ml batch reactor equipped with a spinning basket stirrer (1000 rpm)
- 1 g hardwood (beech) EOL from the Fraunhofer process in 50 ml EtOH
- 100 mg 3 wt% Pd/C catalyst (Degussa) with the particle size of 0.30–0.42 mm
- 200–400 °C, 10–50 bar H_2 (cold pressure), reaction time up to 3 h
- Analysis methods: GPC, elemental analysis, GC/MS and pyrolysis-GC/MS

Experiment Condition Optimisation

Effect of the Temperature



- With increased temperature more depolymerisation and light compounds, and less char, however more volatile/ gaseous products
- Product distribution
  - 200 ºC: much less mono-aromatic products and more unsaturated and bigger compounds compared to 300 ºC
  - 400 ºC: monoaromatic compound distribution much broader compared to 300 ºC

Effect of the H_2 Amount

- With increased pressure less char and more light compounds
- The same products but different ratios

Effect of the Reaction Time

- Results significantly better in the experiment (300 °C, 10 bar H_2) stopped after reaching the reaction temperature than in longer experiments

Best Conditions for the Production of Light Compounds

- 400 ºC and 50 bar (1 h)
- Could still be optimised (reaction time, temperature)

Conclusions

- Hydrotreatment of EOL on the noble metal catalysts has a huge potential
- More catalyst and process development is required

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